



Retrieval Architecture Benchmark: Baseline vs. Hybrid

Executive Summary

As we upgrade our AI infrastructure, a critical decision is how we retrieve company documents to answer user queries. If our retrieval system feeds the wrong documents to our Large Language Model (LLM), the AI will hallucinate or provide incorrect answers.

We conducted a benchmark comparing our legacy retrieval method (**Hashing / Exact Keyword Match**) against a modern **Hybrid Retrieval** pipeline (combining keyword matching with AI-powered semantic understanding and a dedicated "Reranker").

The Bottom Line: The Hybrid approach outperformed the legacy system by **500%** in finding the correct information, proving it is essential for an accurate, reliable, and user-friendly AI product.

The Contenders

- **Baseline: Hashing Vectorizer (Sparse Retrieval)**
 - *How it works:* It looks for exact word matches. If a user searches for "automobile", it only looks for the word "automobile".
 - *The Problem:* It fails completely if a user asks a question using different vocabulary than the document (e.g., searching for "car" when the document says "automobile").
- **Challenger: Hybrid Retrieval + Reranker (Dense + Sparse)**
 - *How it works:* It uses a lightweight AI to understand the *meaning* and *context* of a question, combining it with exact keyword matching. It then uses a "Reranker" as a final judge to push the absolute best answers to the top of the list.

- *The Advantage:* It understands synonyms, concepts, and complex phrasing.

Understanding the Metrics (A Layman's Guide)

To evaluate these systems, we asked 20 highly complex questions based on our data. We told the systems to return their **Top 3** best guesses for the paragraph that answers the question.

Here is how we scored them:

1. Recall@3 (Hit Rate) - *"Did we find the needle in the haystack?"*

- **What it means:** Out of the 20 questions asked, how often did the correct paragraph appear *anywhere* in the Top 3 results?
- **Why it matters to the business:** If the Hit Rate is low, our AI has no chance of answering the user's question correctly because it wasn't given the right data. A high Hit Rate means safety and reliability.

2. Precision@3 - *"How much junk did we bring back?"*

- **What it means:** When we pull 3 results, what percentage of them are actually the correct answer?
- **Why it matters to the business:** Because there is only *one* correct paragraph per question, a "perfect" score here is 0.333 (1 correct out of 3 total). Higher precision means we are feeding the LLM less "garbage" data, which reduces cloud computing costs (fewer tokens) and prevents the AI from getting confused.

3. Mean Reciprocal Rank (MRR) - *"Is the best answer at the very top?"*

- **What it means:** It's not enough to just find the document; we want it to be result #1. MRR gives the system 100 points if the answer is at Rank 1, 50 points for Rank 2, and 33 points for Rank 3.
- **Why it matters to the business:** AI models suffer from "attention fatigue"—they focus heavily on the very first piece of text they read. MRR proves whether our system is putting the most vital information right at the top where the AI will see it.

4. NDCG@3 (Normalized Discounted Cumulative Gain) - *"Overall Ranking Quality"*

- **What it means:** This is the gold-standard metric used by Google and Amazon. It grades the overall "usefulness" of the search results, penalizing the system heavily if it buries the right answer at the bottom of the pile.
- **Why it matters to the business:** It gives us a single, reliable health score for our search engine's intelligence.

Benchmark Results

We ran 20 diverse queries (including exact factoids, vocabulary mismatches, and conceptual questions) against both systems.

Metric	Hashing (Legacy)	Hybrid (New)	Improvement
Recall@3 (Hit Rate)	10.0%	50.0%	5x Better
Precision@3	0.033	0.167	5x Better
Avg MRR	0.100	0.375	3.7x Better
Avg NDCG@3	0.100	0.408	4x Better

(Note: A Precision score of 0.167 means the system retrieved the correct document exactly 50% of the time, as the absolute maximum possible precision score is 0.333).

Business Conclusion: Why Hybrid Wins

The data makes the business case clear. Relying on our legacy Hashing system means our AI will confidently fail **90% of the time** on complex, real-world user queries because it cannot find the right documents.

By upgrading to the Hybrid approach, we achieved a **500% improvement in Hit Rate**.

Key Takeaways for Leadership:

1. **Handles Human Phrasing:** The Hybrid model successfully caught queries where the user phrased things differently than the official documents (e.g., asking about "non-handicapped" instead of "without disabilities").

2. **Saves LLM Costs:** The drastically higher MRR (0.375 vs 0.100) and NDCG scores prove the Hybrid model is pushing the correct answer to the #1 spot. This means we can send less text to the LLM to get a good answer, saving on API costs.
3. **Foundation for Production:** While 50% Recall is a massive leap over the baseline 10%, we can push this even closer to 90%+ in production by fine-tuning the embedding models on our specific company jargon and increasing the final retrieval pool.

Recommendation: Proceed with the deprecation of sparse-only vectorization and fully adopt the Hybrid + Reranker pipeline for our core AI retrieval architecture.